

Prop modulo constante imp constante

Proposición 1. Sea $\Omega \subset \mathbb{C}$ un dominio y $f \in \mathcal{H}(\Omega)$. Entonces

$$|f| \text{ es constante en } \Omega \implies f \text{ es constante en } \Omega.$$

Demostración: Supongamos que $\forall z \in \Omega : |f(z)| = c$ para una constante $c \in \mathbb{R}_{\geq 0}$.

- Si $c = 0$, entonces $f(z) = 0$ para todo $z \in \Omega$, luego f es constante.
- Si $c > 0$, entonces $|f|$ alcanza su máximo c en cualquier punto de Ω . Por el principio del módulo máximo local (~~teo-modulo-maximo~~/Teorema 1), f es constante en Ω . ■

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